

Create and activate a docker Account

- ❖ Docker home page at <https://www.docker.com>.



- ❖ On the Create an docker account page <https://hub.docker.com/signup>, enter your HKU email address (educators/ teaching staff: UID@hku.hk; students: UID@connect.hku.hk), password (a new password at your choice for login to docker) and account name (please put in your Portal UID unless it has been taken up by others), and then click Sign up.

A screenshot of the 'Create your account' page on Docker Hub. It features the Docker logo at the top. Below the heading, there are two buttons: 'Continue with Google' and 'Continue with GitHub'. An 'OR' separator follows. There are three input fields: 'Email', 'Username', and 'Password' (with a toggle for visibility). A checkbox option 'Send me occasional product updates and announcements.' is present. At the bottom, there is a 'Sign up' button and a footer note: 'This site is protected by reCAPTCHA and the Google Privacy Policy and Terms of Service apply.'

Docker for Linux: Setting Up

❖ Introduction

This tutorial is designed to help students to leverage **Docker**, a powerful platform for developing, shipping, and running applications. Docker simplifies the management of application processes in containers, which are isolated from each other and the host system. A **container** is a lightweight, stand-alone, executable package that includes everything needed to run a piece of software, including the code, runtime, libraries, and system settings.

Before diving into the setup of **MongoDB** and **Node-RED**, let's cover the basics of creating a Docker container.

This guidance is applicable only to **Ubuntu** and **macOS**; adjustments may be necessary for Windows.

Additionally, if using a virtual machine for MongoDB, the CPU may need to support hardware virtualization.



Creating Your Docker Container

❖ Installation of Docker

Execute the following commands sequentially:

1. Download the Docker installation script:

```
bash
```

```
curl -fsSL https://test.docker.com -o test-docker.sh
```

2. Run the installation script:

```
bash
```

```
sudo sh test-docker.sh
```



Creating Your Docker Container

- ❖ To test if Docker has been installed successfully, enter the command:

```
sudo docker run hello-world
```

If you see a message similar to the one depicted below, Docker has been successfully installed.

```
nhh@ubuntu:~/nhh$ sudo docker run hello-world
[sudo] password for nhh:
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
c1ec31eb5944: Pull complete
Digest: sha256:4bd78111b6914a99dbc560e6a20eab57ff6655aea4a80c50b0c5491968cbc2e6
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
   (amd64)
3. The Docker daemon created a new container from that image which runs the
   executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it
   to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/

nhh@ubuntu:~/nhh$
```



Install Node-RED Using Docker

❖ 1. Searching for the Node-RED Image Version:

To find the version of the Node-RED image you want to install, enter the command:

```
sudo docker search nodered
```

From the results, we select **nodered/node-red**.

```
nhh@ubuntu:~/nhh$ sudo docker search nodered
```

NAME	DESCRIPTION	STARS	OFFICIAL	AUTOMATED
nodered/node-red	Low-code programming for event-driven applic...	661		
nodered/node-red-docker	Deprecated - older Node-RED Docker images.	361		[OK]
nodered/node-red-dev	Dev/Test builds for Node-RED project (NOT st...	7		
mukowman/nodered-nbrowser-docker	nodered-nbrowser-docker	0		[OK]
marcelocorreia/nodered		1		
asemkhatib/nodered-modified		0		
spants/nodered	Nodered docker built for unraid using Phusio...	5		
openiap/nodered		0		
majortwip/nodered-openhab-bool-calc		0		
trapie/nodered	nodered custom	0		
icebear8/nodered	Node-RED (https://nodered.org/) with git rep...	1		
thiamcheng/nodered-appmetrics		0		
razvoj/nodered		0		
romeokienzler/nodered		0		
burnsys/nodered		0		
chadockerxom/nodered		0		
dennisnotojr/nodered-loadtest	From the NodeRed Base Docker Image, we built...	0		
mbrouwer78/nodered-ha		0		
andrewlaidlaw/nodered-ppc64le		0		
yajrendrag/nodered	clone of https://hub.docker.com/r/nodered/no...	0		
aarlint/noderedfunc		0		
backendbox/noderedapp		0		
redteamsfr/nodered		0		
dianadujing/nodered.edgedemo_amd64		0		
kusold/nodered	nodered with added modules	0		

```
nhh@ubuntu:~/nhh$
```

Install Node-RED Using Docker

❖ 2. Pulling the Image

To pull the image, enter the command: `sudo docker pull nodered/node-red`

```
nhh@ubuntu:~/nhh$ sudo docker pull nodered/node-red
Using default tag: latest
latest: Pulling from nodered/node-red
7264a8db6415: Pull complete
eee371b9ce3f: Pull complete
93b3025fe103: Pull complete
d9059661ce70: Pull complete
485fe505b563: Pull complete
033625439e2c: Pull complete
d4cf1bf4e6f5: Pull complete
4f4fb700ef54: Pull complete
21b718421b0f: Pull complete
31d890393399: Pull complete
2e1de13dbff0: Pull complete
b2a15cef18bc: Pull complete
a1accdfcafe7: Pull complete
bb06094bcfa4: Pull complete
6495b004c211: Pull complete
ac475ea8b303: Pull complete
131fca43d2af: Pull complete
Digest: sha256:75e48924159f6c6bf4221017069a8b496cab8d0b21d3d0b1fcdf058af8294865
Status: Downloaded newer image for nodered/node-red:latest
docker.io/nodered/node-red:latest
nhh@ubuntu:~/nhh$
```

After executing `sudo docker images`, nodered/node-red has been successfully pulled.

```
nhh@ubuntu:~/nhh$ sudo docker images
REPOSITORY          TAG          IMAGE ID      CREATED       SIZE
nodered/node-red    latest      4eb056ef6be0  2 weeks ago  558MB
mysql               5.7         5107333e08a8  6 weeks ago  501MB
ubuntu              latest      174c8c134b2a  6 weeks ago  77.9MB
hello-world         latest      d2c94e258dcf  9 months ago 13.3kB
nhh@ubuntu:~/nhh$
```



Install Node-RED Using Docker

❖ 3. Running the Node-RED Image as a Container

- ① **Create a Directory:** Execute `sudo mkdir /opt/node_red`
- ② **Assign Permissions:** Execute `sudo chmod 777 /opt/node_red`
- ③ **Run the Image:** `sudo docker run -itd -p 1880:1880 -v /opt/node_red:/data --name nodered nodered/node-red`
 - **-v /opt/node_red:/data** mounts the host's /opt/node_red directory to /data in the container.
 - **--name nodered** sets the container's name to nodered. You can see the running container named nodered by executing `sudo docker ps -a`.

```
nhh@ubuntu:~$ sudo docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS
eaba53fa9c03   nodered/node-red  "./entrypoint.sh"       2 minutes ago Up 2 minutes (healthy)  0.0.0.0:1880->1880/tcp, :::1880->1880/tcp
nhh@ubuntu:~$
```

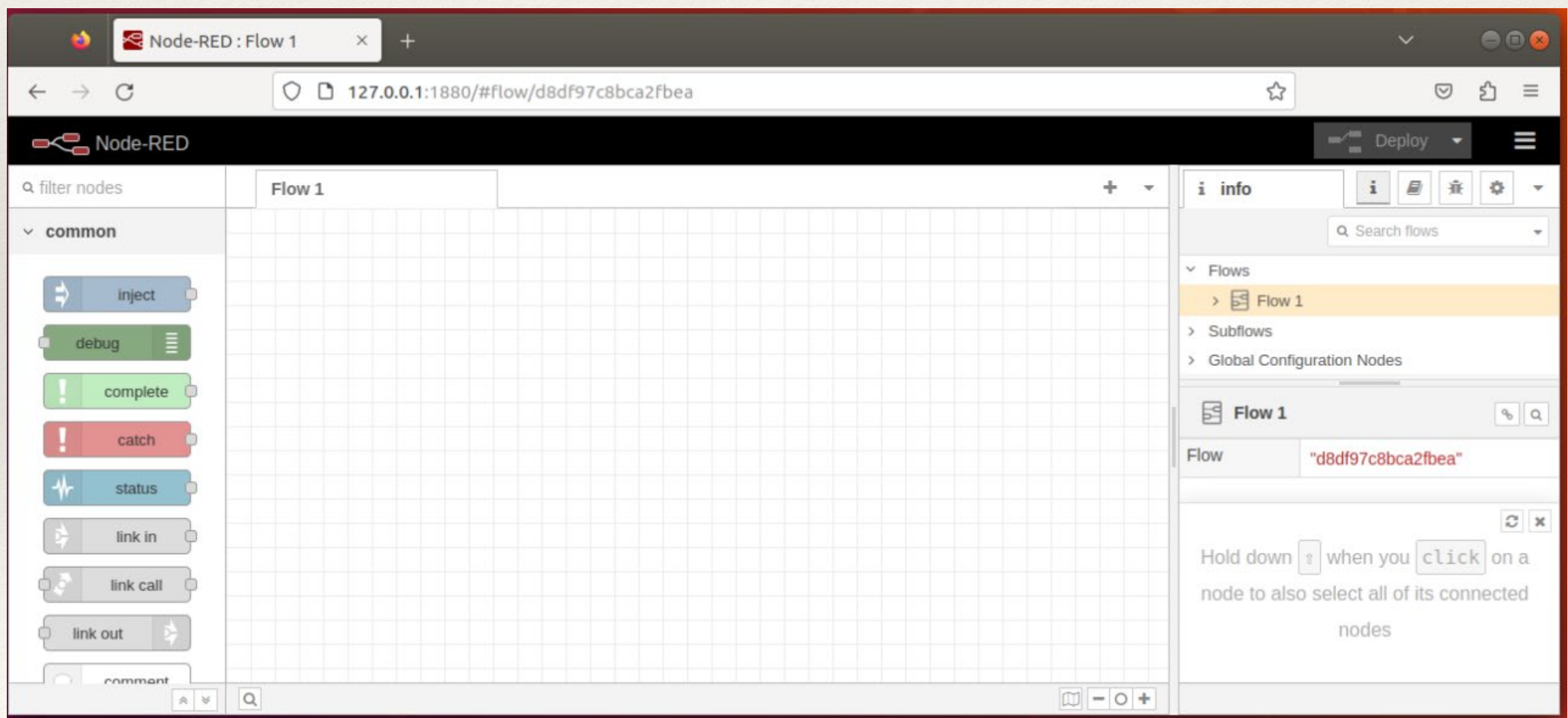
NAMES
nodered



Install Node-RED Using Docker

④ Verifying the Container:

Open Google Chrome or any web browser and enter `http://127.0.0.1:1880`. If you see the Node-RED interface as expected, the container is running correctly.



Install Node-RED Using Docker

❖ 4. Deleting the Container

- ① **Get the Container ID:** Enter `sudo docker ps -a`. Let's say the container ID is `eaba53fa9c03`.

```
nhh@ubuntu:~$ sudo docker ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
eaba53fa9c03	nodered/node-red	"/entrypoint.sh"	9 minutes ago	Up 9 minutes (healthy)	0.0.0.0:1880->1880/tcp, :::1880->1880/tcp	nodered

```
nhh@ubuntu:~$
```

- ② **Stop the Container:** Execute `sudo docker stop eaba53fa9c03`. After this, if you run `sudo docker ps -a`, you should see the container's status as Exited.

```
nhh@ubuntu:~$ sudo docker ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
eaba53fa9c03	nodered/node-red	"/entrypoint.sh"	10 minutes ago	Exited (143) 4 seconds ago		nodered

```
nhh@ubuntu:~$
```

- ③ **Delete the Container:** Execute `sudo docker rm -f eaba53fa9c03`. Afterward, running `sudo docker ps -a` will show that the container no longer exists.

```
nhh@ubuntu:~$ sudo docker ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
--------------	-------	---------	---------	--------	-------	-------

```
nhh@ubuntu:~$
```



Install mongodb Using Docker

❖ 1. Pulling the Image

- ① To download the latest MongoDB image, enter the command: `sudo docker pull mongo:latest`
- ② After executing `sudo docker images`, you will see that the MongoDB image has been successfully pulled.

```
nhh@ubuntu:~$ sudo docker images
```

REPOSITORY	TAG	IMAGE ID	CREATED	SIZE
nodered/node-red	latest	4eb056ef6be0	2 weeks ago	558MB
mongo	latest	7ee26c8012da	3 weeks ago	757MB
mysql	5.7	5107333e08a8	6 weeks ago	501MB
ubuntu	latest	174c8c134b2a	6 weeks ago	77.9MB
hello-world	latest	d2c94e258dcb	9 months ago	13.3kB

```
nhh@ubuntu:~$
```



Install mongodb Using Docker

❖ 2. Running the MongoDB Image as a Container and Verification

① **Create a Directory:** Execute `sudo mkdir /opt/mongodb.`

② **Assign Permissions:** Execute `sudo chmod 777 /opt/mongodb.`

③ **Run the Image:** `sudo docker run -itd -p 27017:27017 -v /opt/mongodb:/data/db --name Mymongo mongo:latest`

- **-v /opt/mongodb:/data/db** mounts the host's /opt/mongodb directory to /data/db in the container.
- **--name Mymongo** sets the container's name to Mymongo. You can see the running container named Mymongo by executing `sudo docker ps -a`.

```
nhh@ubuntu:~$ sudo docker ps -a
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS                               NAMES
a9a2f83985ff   mongo:latest  "docker-entrypoint.s..."  8 seconds ago  Up 7 seconds  0.0.0.0:27017->27017/tcp, :::27017->27017/tcp  Mymongo
```



Install mongodb Using Docker

④ Entering the Container

To enter the container, use the command: `sudo docker exec -it Mymongo /bin/bash`

```
nhh@ubuntu:~$ sudo docker exec -it Mymongo /bin/bash
root@a9a2f83985ff:/#
```

Then, start the MongoDB shell by entering `mongosh`.

```
root@a9a2f83985ff:/# mongosh
Current Mongosh Log ID: 65b8856ed75beb7a63db0b63
Connecting to:      mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.1.1
Using MongoDB:      7.0.5
Using Mongosh:       2.1.1

For mongosh info see: https://docs.mongodb.com/mongosh-shell/

To help improve our products, anonymous usage data is collected and sent to MongoDB periodically (https://www.mongodb.com/legal/privacy-policy).
You can opt-out by running the disableTelemetry() command.

-----
The server generated these startup warnings when booting
2024-01-30T05:05:26.026+00:00: Using the XFS filesystem is strongly recommended with the WiredTiger storage engine. See http://dochub.mongodb.org/core/prodnotes-filesystem
2024-01-30T05:05:26.584+00:00: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
2024-01-30T05:05:26.585+00:00: vm.max_map_count is too low
-----
```



Install mongodb Using Docker

Create an admin user with the following commands:

```
use admin
db.createUser(
{
user: "admin",
pwd: " 666712",
  roles: [ { role: "root", db: "admin" } ]
}
);
```

```
test> use admin
switched to db admin
admin> db.createUser(
... {
... user: "admin",
... pwd: "666712",
... roles: [ { role: "root", db: "admin" } ]
... }
... );
{ ok: 1 }
admin> 
```



Install mongodb Using Docker

- ❖ To exit the MongoDB shell, type exit, then type exit again to leave the container.

```
admin> exit
root@a9a2f83985ff:/# exit
exit
nhh@ubuntu:~$
```

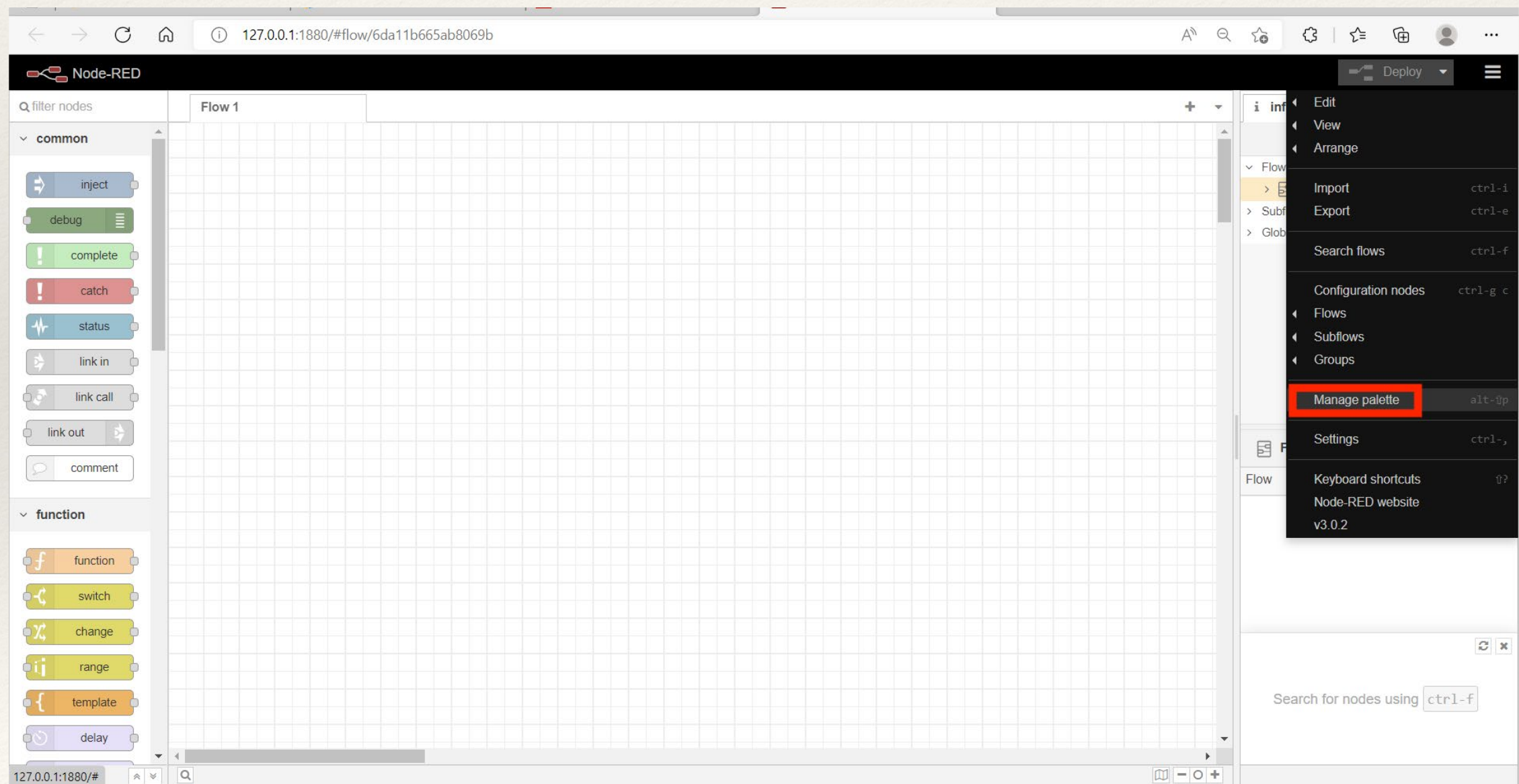
- ❖ To connect to MongoDB within the container, use: `mongo --port 27017 -u admin -p 666712 --authenticationDatabase admin`. After connecting, entering `show dbs` will display the databases, including the admin database you created.

```
nhh@ubuntu:~$ mongo --port 27017 -u admin -p 666712 --authenticationDatabase admin
MongoDB shell version v3.6.3
connecting to: mongodb://127.0.0.1:27017/
MongoDB server version: 7.0.5
WARNING: shell and server versions do not match
Welcome to the MongoDB shell.
For interactive help, type "help".
For more comprehensive documentation, see
  http://docs.mongodb.org/
Questions? Try the support group
  http://groups.google.com/group/mongodb-user
Server has startup warnings:
{"t":{"$date":"2024-01-30T05:05:26.026+00:00"},"s":"I", "c":"STORAGE", "id":22297, "ctx":"initandlisten","msg":"Using the XFS filesystem is strongly recommended with the WiredTiger storage engine. See http://dochub.mongodb.org/core/prodnotes-filesystem","tags":["startupWarnings"]}
{"t":{"$date":"2024-01-30T05:05:26.584+00:00"},"s":"W", "c":"CONTROL", "id":22120, "ctx":"initandlisten","msg":"Access control is not enabled for the database. Read and write access to data and configuration is unrestricted","tags":["startupWarnings"]}
{"t":{"$date":"2024-01-30T05:05:26.585+00:00"},"s":"W", "c":"CONTROL", "id":5123300, "ctx":"initandlisten","msg":"vm.max_map_count is too low","attr":{"currentValue":5530,"recommendedMinimum":1677720,"maxConns":838860},"tags":["startupWarnings"]}
> show dbs
admin    0.000GB
config  0.000GB
local    0.000GB
>
```



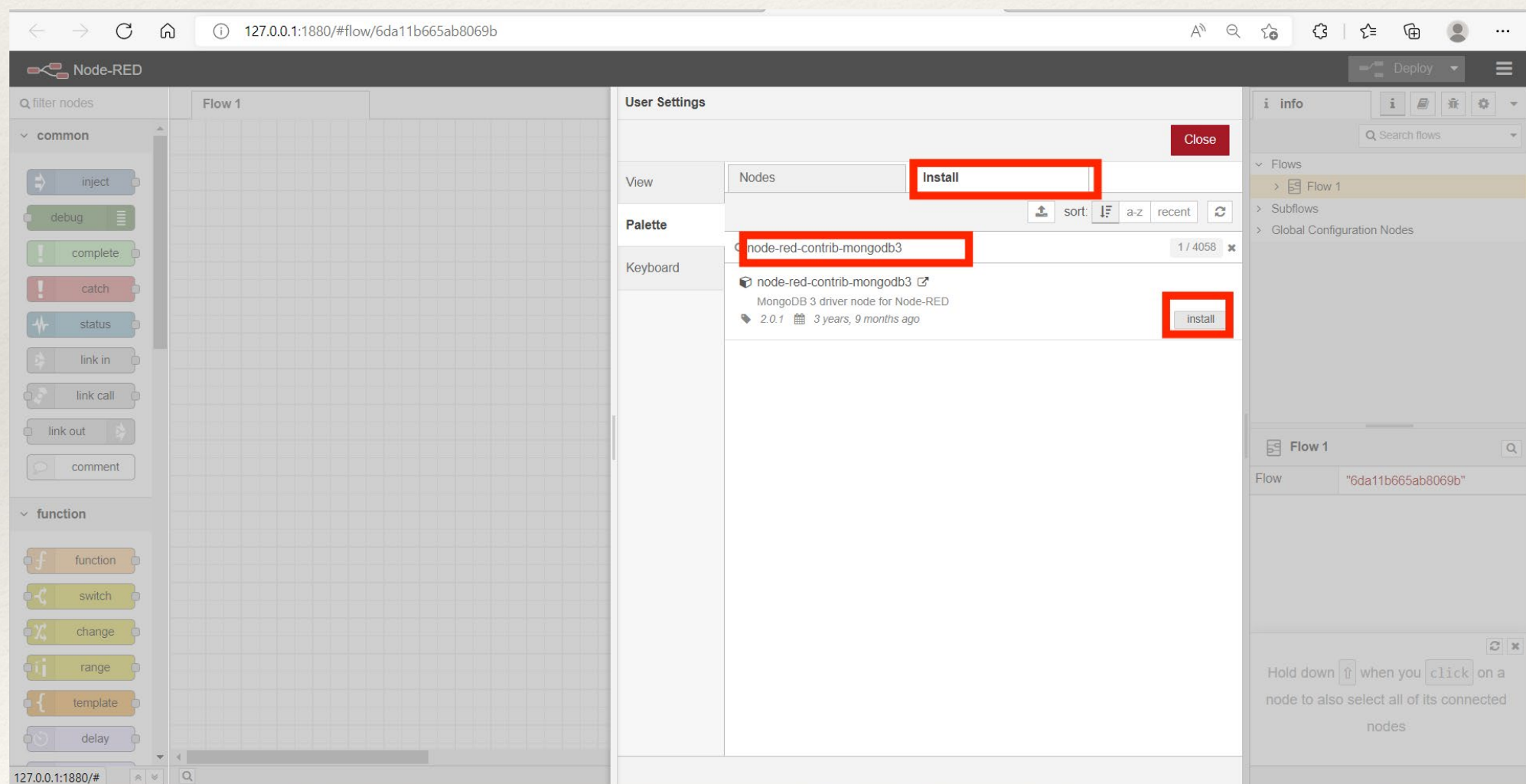
Installing MongoDB and Dashboard Modules in NodeRed

- ❖ Use a browser to access the URL: <http://localhost:1880>
- ❖ Click the top right icon, select and click **Manage Palette**



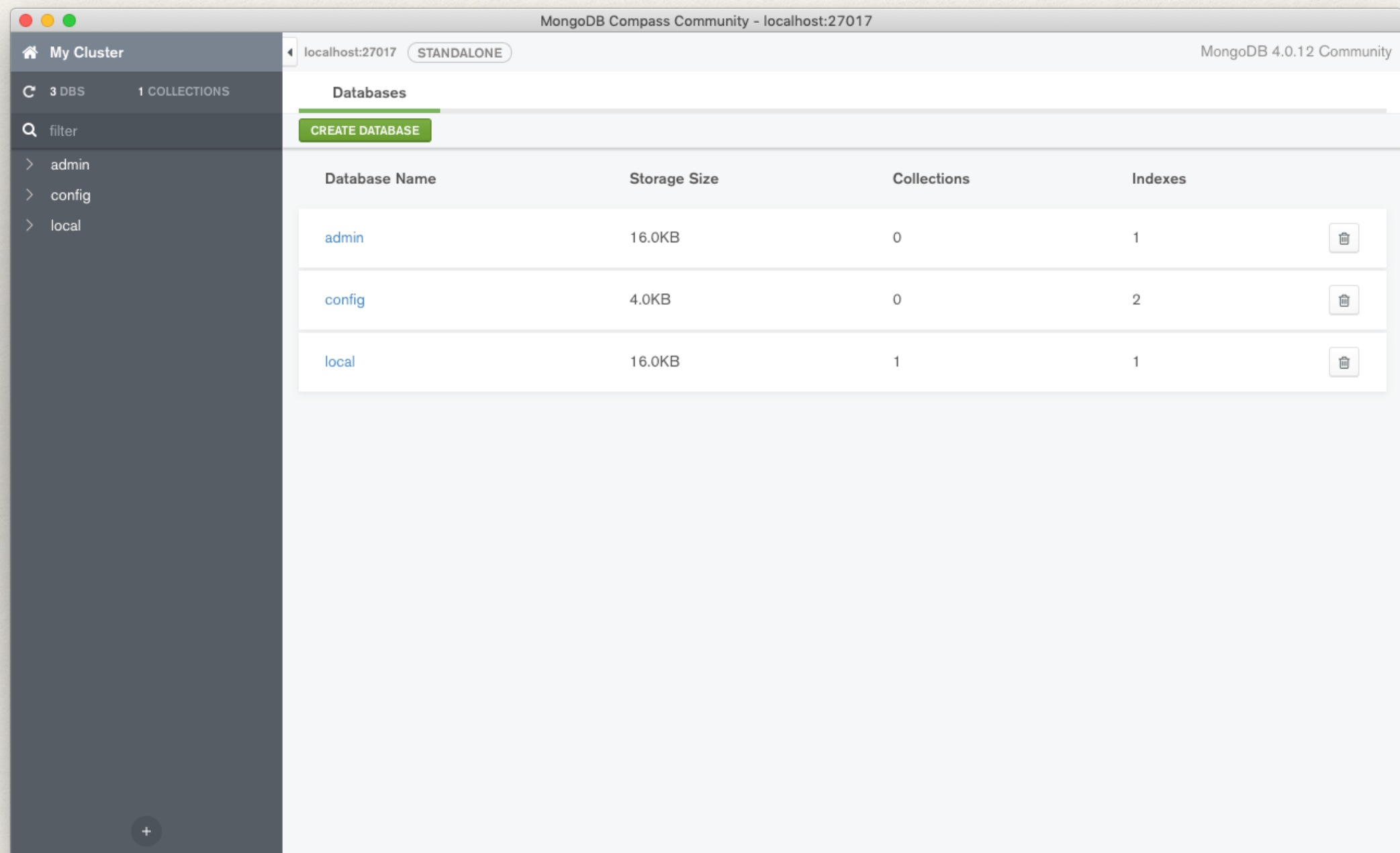
Installing MongoDB and Dashboard Modules in NodeRed

- ❖ Select the Install Tab, and input MongoDB in the search field.
- ❖ Scroll down and locate the row with **node-red-contrib-mongodb3**, and click **Install** button, and click **Install** button again in the pop-up window.



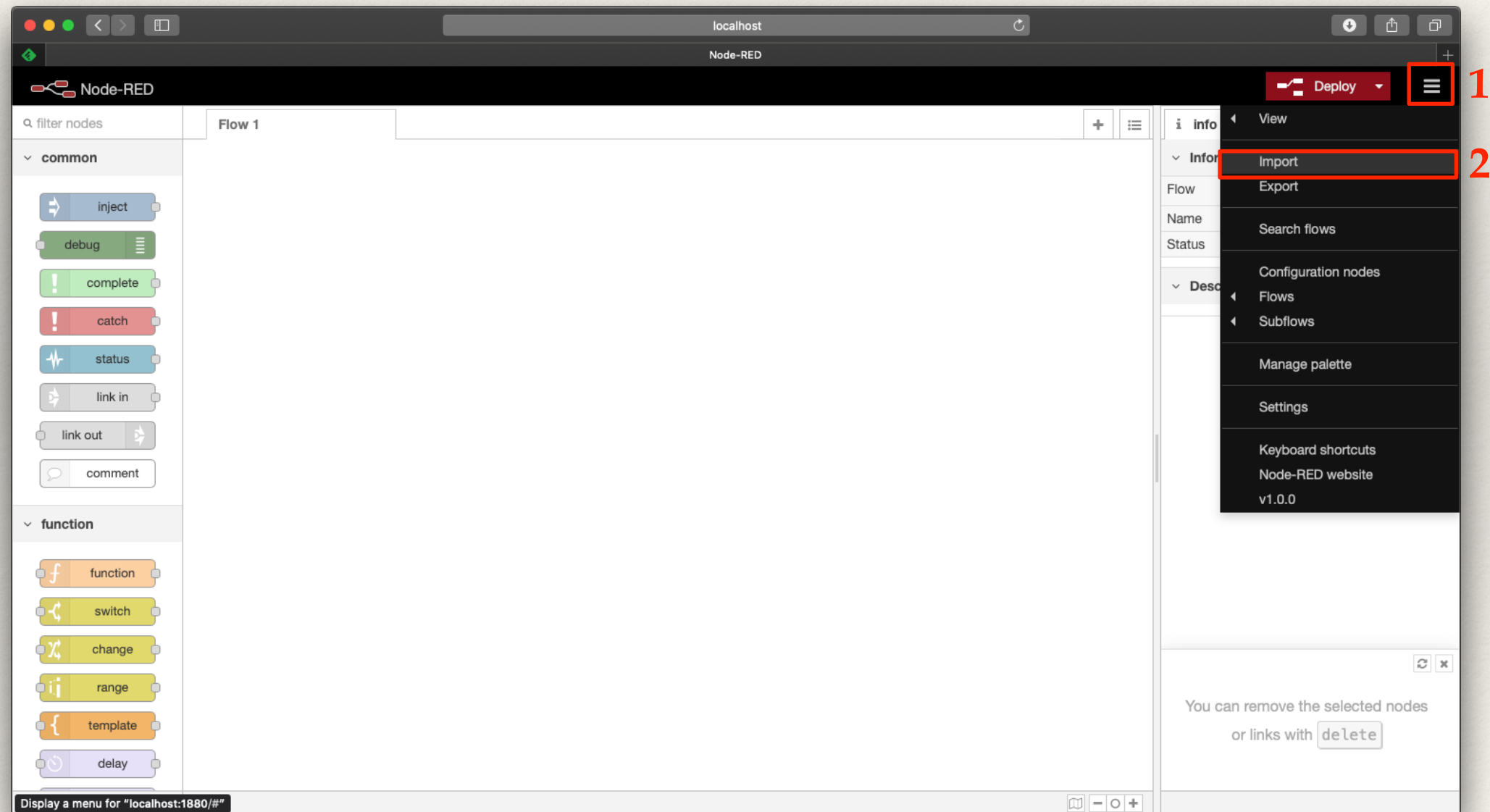
Accessing MongoDB

- ❖ Once connected, you should be able to browse your data stored in your MongoDB instance.



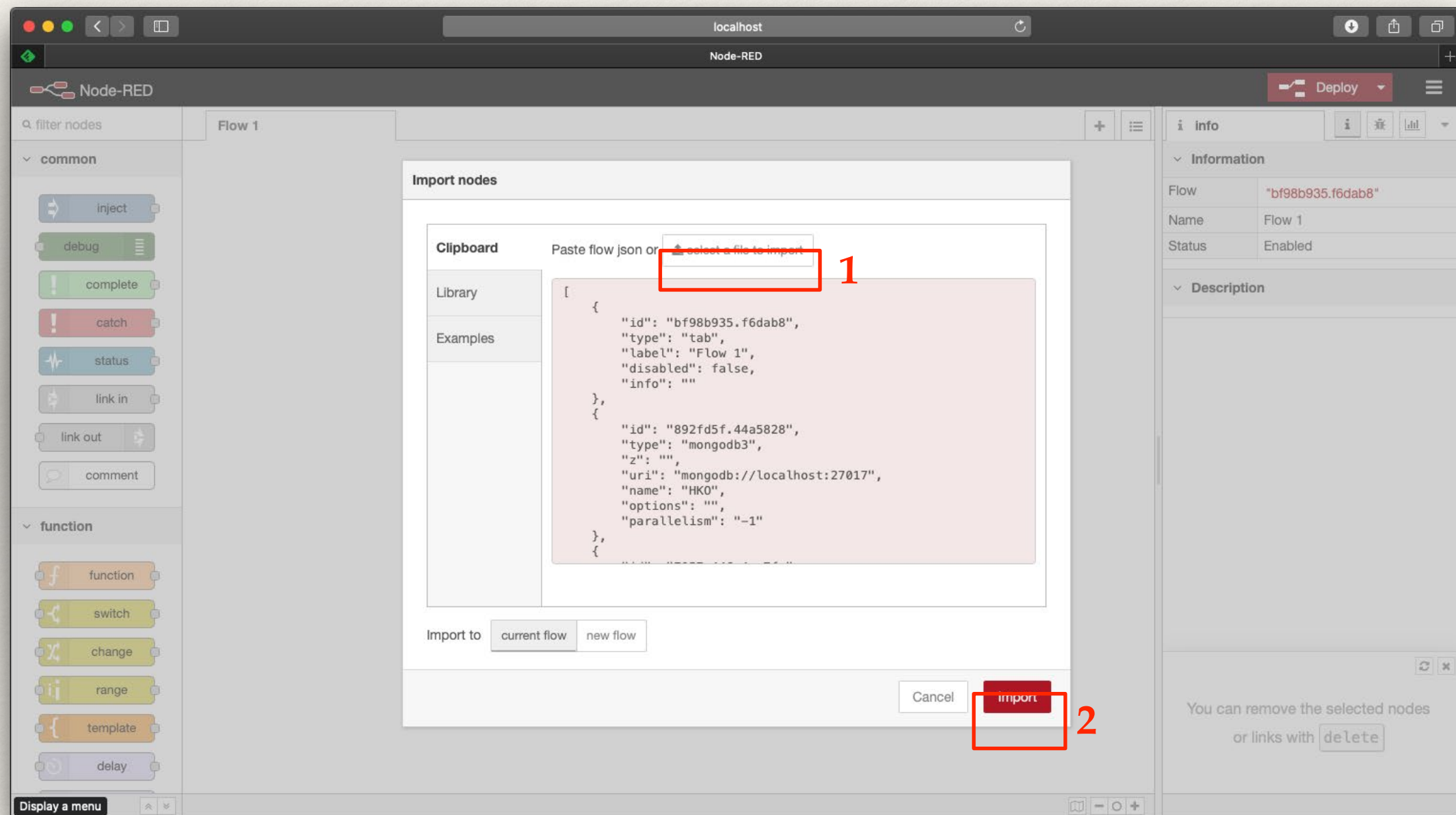
Node-Red Example Program

- ❖ Download the the HKO.Flow.txt file from moodle, and import it into your NodeRed environment:



Node-Red Example Program

- ❖ Download the the HKO.Flow.json file from moodle, and import it into your NodeRed environment:



Node-Red Example Program

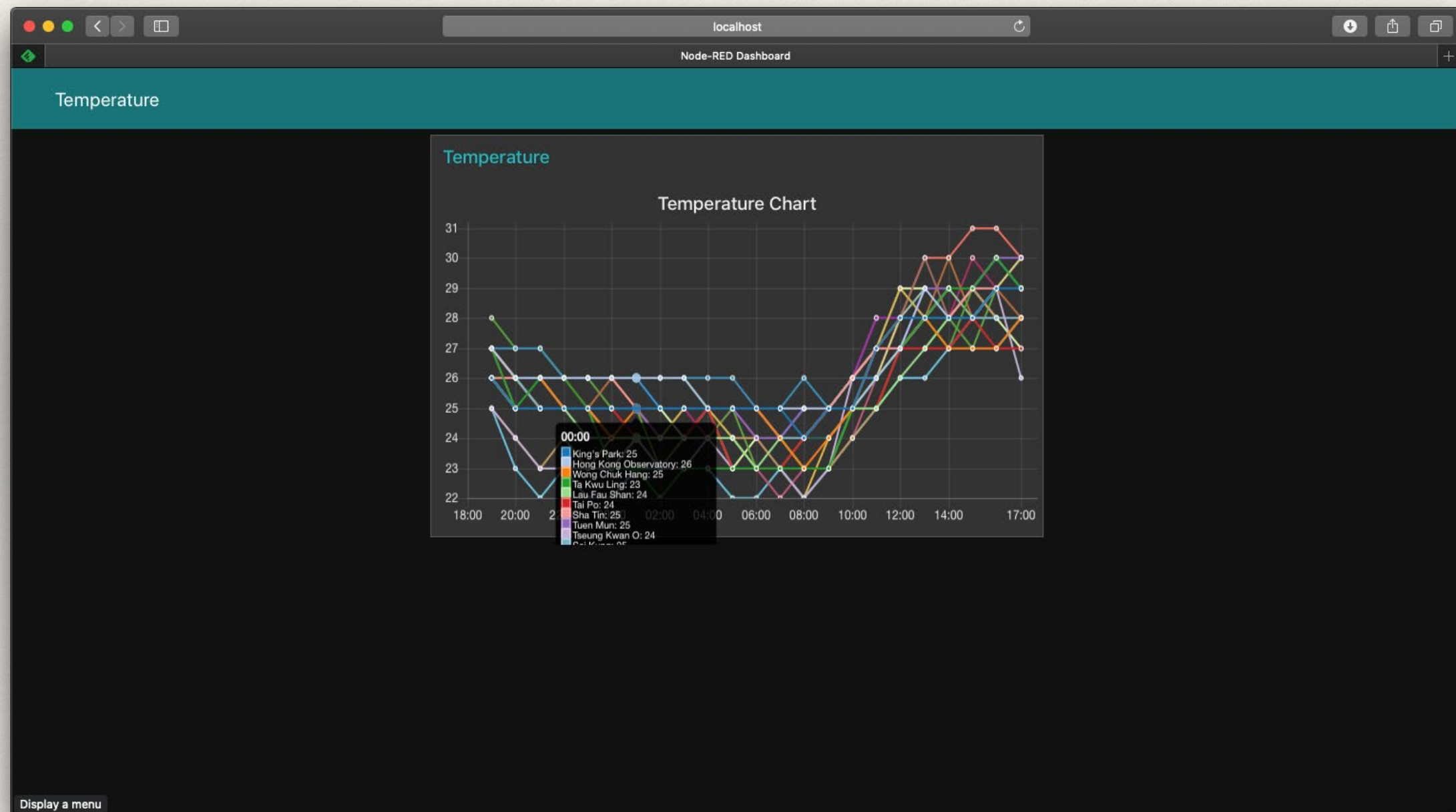
- ❖ After importing the flow, click **Deploy** button

The screenshot displays the Node-RED web interface in a browser window. The main workspace shows a flow named 'Flow 1' with two parallel processing chains. The top chain starts with a 'Timer:timestamp' node, followed by 'HKO Retrieve Last Record', 'Retrieve Last Record Query', 'Get Last updateTime', 'http request', 'json', 'New Record Available', 'Need Update', 'No Update', 'HKO Weather Report insert', and 'msg.payload'. The bottom chain starts with 'Timer:timestamp', followed by 'Construct Queries', 'Construct Date Range', 'HKO Retrieve Records', 'Format Chart Data', 'Temperature Chart', and 'msg.payload'. The right sidebar shows the 'Info' tab with a red box around the 'Deploy' button and the 'Information' section. The 'Information' section shows the flow ID '431c4fd7.cc792', name 'Flow 1', and status 'Enabled'.

Node-RED interface showing a flow with two parallel processing chains. The top chain starts with a 'Timer:timestamp' node, followed by 'HKO Retrieve Last Record', 'Retrieve Last Record Query', 'Get Last updateTime', 'http request', 'json', 'New Record Available', 'Need Update', 'No Update', 'HKO Weather Report insert', and 'msg.payload'. The bottom chain starts with 'Timer:timestamp', followed by 'Construct Queries', 'Construct Date Range', 'HKO Retrieve Records', 'Format Chart Data', 'Temperature Chart', and 'msg.payload'. The right sidebar shows the 'Info' tab with a red box around the 'Deploy' button and the 'Information' section. The 'Information' section shows the flow ID '431c4fd7.cc792', name 'Flow 1', and status 'Enabled'.

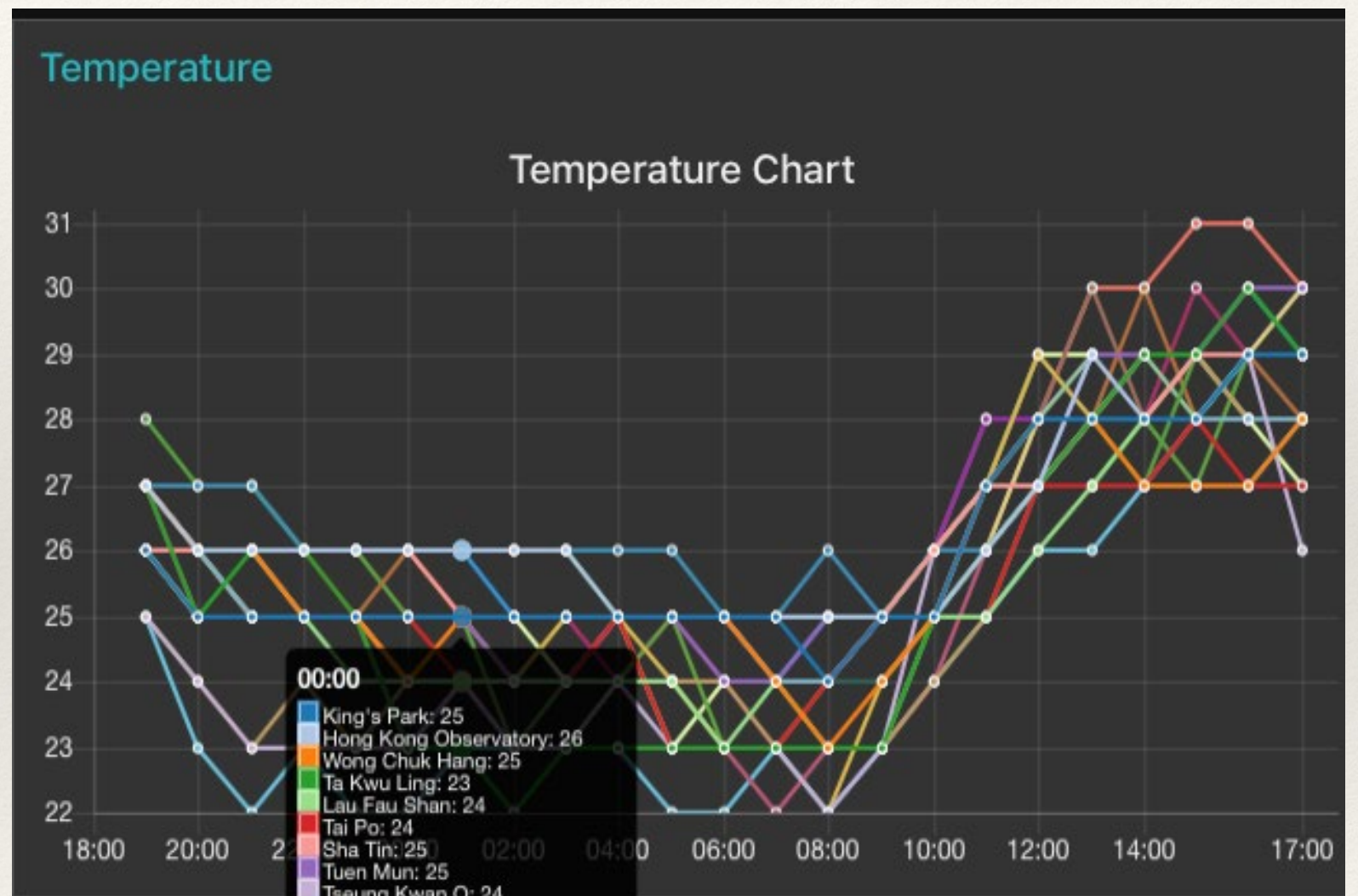
Node-Red Example Program

- ❖ After deployment, you can let the node-red instance to run for a couple of hours, and then launch another browser window and open the URL: <http://localhost:1880/ui>



Node-Red Example Program

- ❖ You should see a chart, which plots the historical temperature of various Hong Kong locations up to the last 24 hours



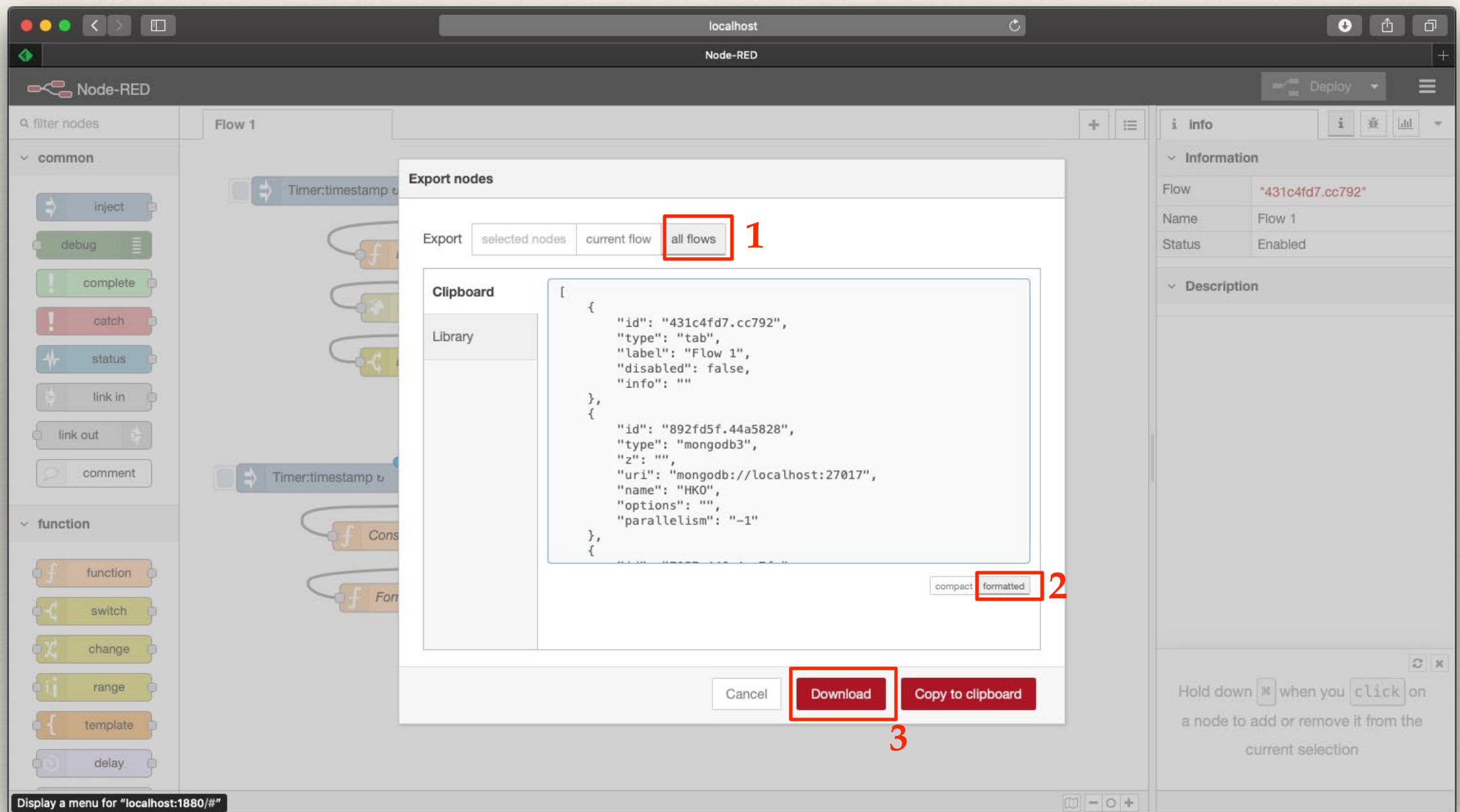
Exporting Node-Red Flow for Submission

- ❖ Click the top right icon, and select **Export**

The screenshot displays the Node-RED web interface in a browser window. The main workspace shows a flow named 'Flow 1' with two parallel processing paths. The top path starts with a 'Timer:timestamp' node, followed by an 'HKO Retrieve Last Record' node (showing 4 successes), a 'Retrieve Last Record Query' function node, a 'Get Last updateTime' function node, an 'http request' node, a 'json' node, a 'New Record Available' function node, a 'Need Update' node, a 'No Update' function node, an 'HKO Weather Report Insert' node, and finally a 'msg.payload' output node. The bottom path starts with another 'Timer:timestamp' node, followed by a 'Construct Queries' function node, a 'Construct Date Range' function node, an 'HKO Retrieve Records' node (showing 4 successes), a 'Format Chart Data' function node, a 'Temperature Chart' node, and a 'msg.payload' output node. On the left, the 'common' and 'function' node palettes are visible. On the right, the 'Deploy' button is highlighted with a red box and labeled '1'. Below it, the menu icon (three horizontal lines) is also highlighted with a red box and labeled '2'. The menu is open, showing options like 'View', 'Import', 'Export', 'Search flows', 'Configuration nodes', 'Flows', 'Subflows', 'Manage palette', 'Settings', 'Keyboard shortcuts', 'Node-RED website', and 'v1.0.0'. The 'Export' option is highlighted with a red box.

Exporting Node-Red Flow for Submission

- ❖ Select all flows, formatted, and click Download button



Assignment Submission

- ❖ After completion of your assignment, you should export your flows in your node-red environment for submission in Moodle.
- ❖ You should also submit provide clear procedure to reproduce your development environment in Moodle as well, so that your tutor can access your docker instance for grading purpose.

